

User's Manual

DIELECTRIC CONSTANT OF SOLIDS & LIQUIDS

Model: DSL-01

(Rev:3-3-22)

Manufactured by:

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ISO 9001:2015
Certified Company

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COPYRIGHT AND WARRANTY

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LIMITED WARRANTY

SES Instruments Pvt. Ltd warrants this product to be free from defects in materials and workmanship for a period of one year from the date of shipment to the customer. SES Instruments Pvt. Ltd will repair or replace, at its option, any part of the product which is deemed to be defective in material or workmanship. This warranty does not cover damage to the product caused by abuse or improper use. Determination of whether a product failure is the result of manufacturing defect or improper use by the customer shall be made solely by SES Instruments Pvt. Ltd. Responsibility for the return of equipment for warranty repair belongs to the customer. Equipment must be properly packed to prevent damage and shipped postage or freight prepaid. (Damage caused by improper packaging of the equipment for return shipment will not be covered by the warranty). Shipping costs for returning the equipment, after repair, will be paid by SES Instruments Pvt. Ltd.

EQUIPMENT RETURN

Should this product have to be returned to SES Instruments Pvt. Ltd, for whatever reason, notify SES Instruments Pvt. Ltd BEFORE returning the product. Upon notification, the return authorization and shipping instructions will be promptly issued.

Note : No equipment will be accepted for return without an authorization.

When returning equipment for repair, the units must be packed properly. Carriers will not accept responsibility for damage by improper packing. To be certain the unit will not be damaged in shipment, observe the following rules:

1. The carton must be strong enough for the item shipped.
2. Make certain there is at least two inches of packing material between any point on the apparatus and the inside walls of the carton.
3. Make certain that the packing material can not displace in the box, or get compressed, thus letting the instrument come in contact with the edge of the box.

SAFETY INFORMATION

This Section addresses safety considerations and describes symbols that may appear on the Instrument or in the manual.

A **Warning** Statement identifies conditions or practices that could result in injury or death. A **Caution** statement identifies conditions or practices that could result in damage to the Instrument or equipment to which it is connected.



To avoid electric shock, personal injury, or death, carefully read the information in Table-1, “Safety Information,” before attempting to install, use, or service the Instrument.

GENERAL SAFETY SUMMARY

This equipment is Class 1 equipment tested in accordance with the European Standard publication EN 61010-1.

This manual contains information and warnings that must be observed to keep the Instrument in a safe condition and ensure safe operation.

To use the Instrument correctly and safely, read and follow the precautions in Table 1 and follow all safety instructions or warnings given throughout this manual that relate to specific measurement functions. In addition, follow all generally accepted safety practices and procedures required when working with and around electricity.

SYMBOLS

Table 2 lists safety and electrical symbols that appear on the Instrument or in this manual.

Table 2. Safety and Electrical Symbols

Symbols	Description	Symbols	Description
	Risk of danger. Important information. See Manual.		Earth ground
	Hazardous voltage. Voltage >30Vdc or ac peak might be present.		Potentially hazardous voltage
	Static awareness. Static discharge can damage parts.		Do not dispose of this product as unsorted municipal waste. Contact SES or a qualified recycle for disposal.

Table 1. Safety Information



To avoid possible electric shock, personal injury, or death, read the following before using the Instrument:

- **Use the Instrument only as specified in this manual, or the protection provided by the Instrument might be impaired.**
- **Do not use the Instrument in wet environments**
- **Inspect the Instrument in wet environments.**
- **Inspect the Instrument before using it. Do not use the Instrument if it appears damaged.**
- **Inspect the connecting lead before use. Do not use them if insulation is damaged or metal is exposed. Check the connecting leads for continuity. Replace damaged connecting leads before using the Instrument.**
- **Whenever it is likely that safety protection has been impaired, make the Instrument inoperative and secure it against any unintended operation.**
- **Have the Instrument serviced only by qualified service personnel.**
- **Always use the power cord and connector appropriate for the voltage and outlet of the country or location in which you are working.**
- **Never remove the cover or open the case of the Instrument before without first removing it from the main power source.**
- **Never operate the Instrument with the cover removed or the case open.**
- **Use only the replacement fuses specified by the manual.**
- **Do not operate the Instrument around explosive gas, vapor or dust.**
- **When servicing the Instrument, use only specified replacement parts.**
- **The equipment can remain Switched on continuously for five hours**
- **The equipment must remain Switched off for at least fifteen minutes before being switched on again.**
- **The equipment is only for the intended use**
- **Use the equipment only as specified in this manual.**

Unpacking and Inspecting the Instrument

Every care is taken in the choice of packing material to ensure that your Instrument will reach you in perfect condition. If the Instrument has been subject to excessive handling in transit, there may be visible external damage to the shipping container and packing material for the carrier's inspection.

Carefully unpack the Instrument from its shipping container and inspect the contents for damaged or missing items. If the Instrument appears damaged or something is missing, contacts the carrier and SES immediately. Save the container and packing material in case you have to return the Instrument.

Storing and Shipping the Instrument

To prepare the Instrument for storage or shipping, if possible, use the original shipping container alongwith thermocoal corners, as it provides shock isolation for normal handling operations. If the original shipping container is not available, use any good cardboard box which is at least 2-3 inches bigger than the instrument on all sides, with cushioning material (thermocoal or styrofoam etc) that fills the space between the Instrument and the side of this box.

To store the Instrument, place the box under cover in a location that complies with the storage environment specification described in the “Environment Sections” below.

Environment

Temperature

Operating 0°C to 50°C

Storage 40°C to 70°C

Warm Up15 min to full uncertainty specification

Relatively Humidity (non-condensing)

OperatingUncontrolled (<10°C)

<90 % (10°C to 30°C)

<75 % (30°C to 40°C)

<45 % (40°C to 50°C)

Storage..... -10°C to 60°C <95 %

Power Considerations

The Instrument operates on varying power distribution standards found throughout the world and must be set up to operate on the line voltage that will power it. The Instrument is packed ready for use with a line voltage determined at the time of ordering.

Replacing the Fuses

The Instrument uses one fuse to protect the line-power input and two fuses to protect current-measurement inputs.

Line-Power Fuse

The Instrument has a line-power fuse in series with the power supply. Table 3 indicates the proper fuse for each of the four line-voltage selections. The line-power fuse is accessed through the rear panel.

1. Unplug the power cord.
2. Rotate the fuse holder cap to the right until the fuse POPS out.
3. Remove the fuse and replace it with a fuse of an appropriate rating for the selected line-power voltage. See Table 2.



Warning

To avoid electric shock or fire, do not use makeshift fuses or short-circuit the fuse holder.

Table 2. Line Voltage to Fuse Rating

Line Voltage Selection	Fuse Rating
220/ 240 V	1A, 250V (Slow blow)
100/ 120 V	2A, 250 V (Slow blow)

Connecting to Line Power



Warning

To avoid shock hazard, connect the factory supplies three conductor line power cord to a properly grounded power outlet. Do not use a two-conductor adapter or extension cord, as this will break the protective ground connection. If a two conductor power cord must be used, a protective grounding wire must be connected between the ground terminal and earth ground before connecting the power cord or operating the Instrument.

1. Verify that the Line voltage is set to the correct setting.
2. Verify that the correct fuse for the line voltage is installed.
3. Connect the power cord to a properly grounded three-prong outlet. See Figure 1 for line-power cord types available from SES. Refer to Table 2 for description of the line-power cords.

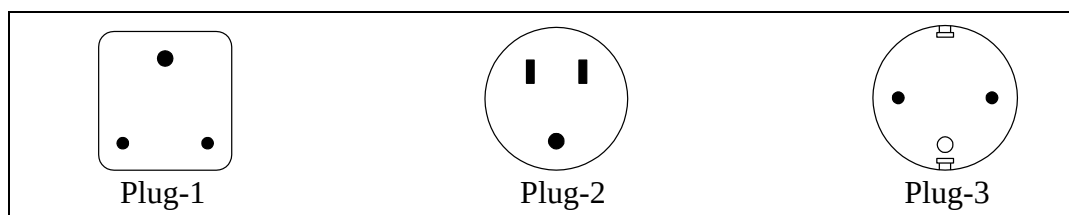


Figure 1. Line-Power Cord Types Available from SES

Table 3. Line-Power Cord Types Available from SES

Type	Voltage/Current	SES Model Number
India	240 V/ 5 A	Plug-1
North America	120 V/15 A	Plug-2
Universal Euro	220 V/16 A	Plug-3

Turning Power On

The On-Off switch on the front panel when points towards “ON” signs, indicates that the equipment has been switched on.

Cleaning the Instrument

 Warning

To avoid electric shock or damage to the Instrument, never get water inside the Instrument.

 Caution

To avoid damaging the Instrument’s housing, do not apply solvents to the Instrument. If the Instrument requires cleaning, wipe it down with a cloth that is lightly dampened with water or a mild detergent. Do not use aromatic hydrocarbons, alcohol, chlorinated solvents, or methanol-based fluids when wiping down the Instrument.

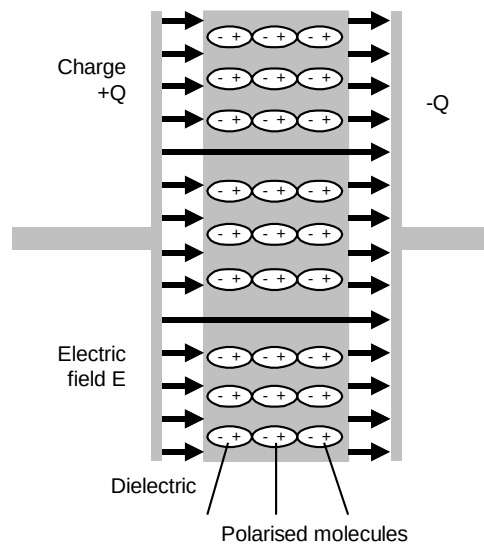


Figure 1

INTRODUCTION

Dielectric or electrical insulating materials are the substances in which electrostatic field can persist for long times. When a dielectric is placed between the plates of a capacitor and the capacitor is charged, the electric field between the plates polarizes the molecules of the dielectric (Figure 1). This produces concentration of charge on its surface that creates an electric field which is antiparallel to the original field (which has polarized the dielectric). This reduces the electric potential difference between the plates. Considered in reverse, this means that, with a dielectric between the plates of a capacitor, it can hold a larger charge. The extent of this effect depends on the dipole polarizability of molecules of the dielectric, which in turn determines the dielectric constant of the material.

The electrons in the molecules shift toward the positively charged plate. The molecules then create a leftward electric field that partially annuls the field created by the charged plates. (The air gap is shown for clarity; in a real capacitor, the dielectric is in direct contact with the plates.)

The method for determination of dielectric constants of liquids consists in the successive measurement of capacitance, first in a vacuum, and then when the capacitor is immersed in the liquid under investigation. A cylindrical capacitor has been used for liquid samples and different size parallel plate capacitors for solid samples.

Parallel Plate Capacitor:

The capacity of a parallel plate capacitor is given by

$$C = \kappa \frac{\epsilon_0 A}{d} \text{ farad,} \quad (1)$$

where A is plate area and d is distance between the plates. It is assumed that the dielectric completely fills the space between the capacitor plates.

Cylindrical Capacitor:

The capacitance per unit length of a long cylindrical capacitor immersed in a medium of dielectric constant k is given by

$$C' = \kappa \frac{2\pi\epsilon_0}{\ln(r_2/r_1)} \text{ farad/m.} \quad (2)$$

Here $\epsilon_0 = 8.854 \times 10^{-12} \text{ C}^2\text{m}^{-2}\text{N}^{-1}$ is permittivity of free space, r_1 is external radius of the inner cylinder and r_2 is internal radius of the outer cylinder.

In actual practice, there are errors due to stray capacitances at the ends of the cylinders and the leads. In any accurate measurement, it is necessary to eliminate these. It has been done in the following way:

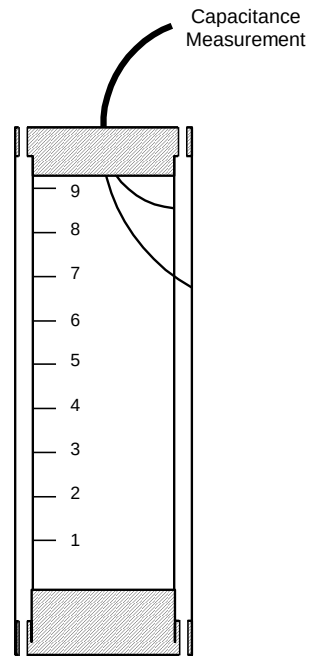


Figure 2a: Coaxial Cylinder Set-up (CCS)

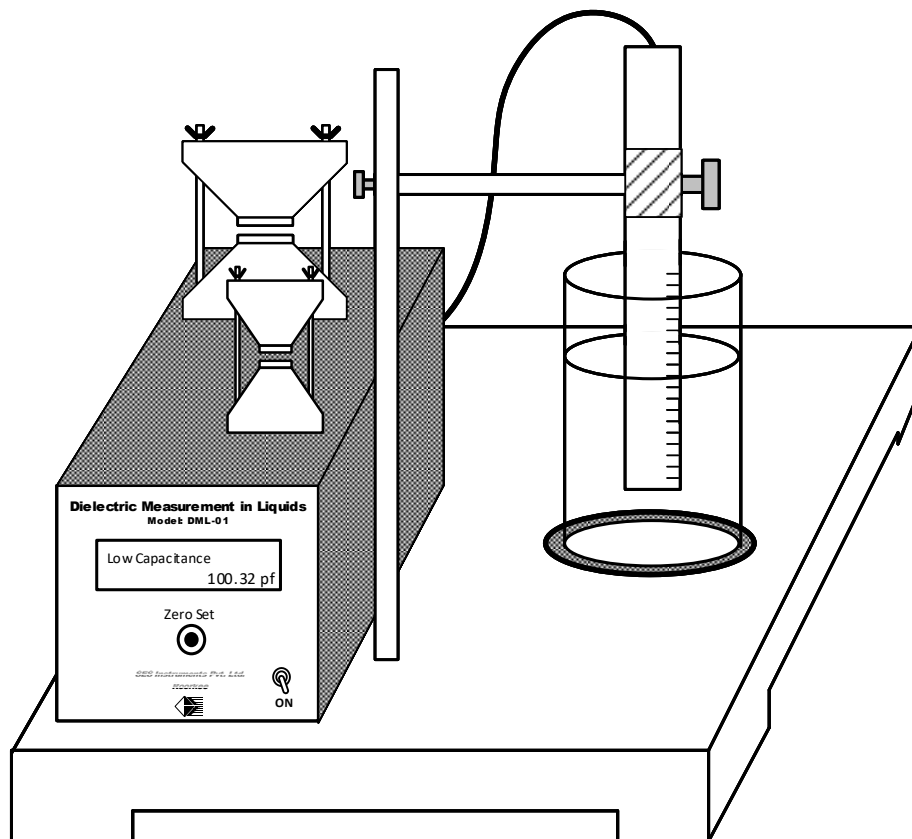


Figure 2b
Dielectric Measurement in Solids & Liquids (Non Conducting)
Model: DSL-01

Consider a cylindrical capacitor of length L (in meter) filled to a height $h < L$ with a liquid of dielectric constant κ . Its total capacitance is given by

$$C = \frac{2\pi\epsilon_0}{\ln(r_2/r_1)} [\kappa h + \kappa_0 (L - h)] + C_s. \quad (3)$$

Here κ_0 is the dielectric constant of air and C_s is sum total of stray capacitances. A simplification of Eq.(2) leads to

$$C = \frac{2\pi\epsilon_0}{\ln(r_2/r_1)} (\kappa - \kappa_0) h + C_0, \quad (4)$$

where

$$C_0 = C_s + \frac{2\pi\epsilon_0}{\ln(r_2/r_1)} \kappa_0 L.$$

Eq. (3) shows that the measured capacity C is a linear function of h (the height upto which the liquid is filled in the capacitor). If we vary the liquid height h , and measure it, together with the corresponding capacitance C , the plot of the data should be a straight line. The slope

$$\frac{dC}{dh} = \frac{2\pi\epsilon_0}{\ln(r_2/r_1)} (\kappa - \kappa_0) \text{ farad/m} \quad (5)$$

will yield a value for κ if κ_0 , r_1 and r_2 are known. The uncertainty due to C_s has thus been eliminated.

PACKING LIST

1. Dielectric Constant of Solids & Liquids, DSL-01: One
2. Sample Jar (Glass): One
3. Sample (Solid): Glass Plate, Bakelite Sheet, Teflon, PZT (Lead Zirconate Titanate)
4. Sample (Liquid): Carbon Tetrachloride (500ml)

BRIEF DESCRIPTION OF THE APPARATUS

1. (a) Probes Arrangement for Solids

It consists of two circular parallel plate setups of diameters 10mm and 50mm to cater to smaller and bigger samples respectively. One of the plates of the pair is fixed on the mount while the other can be moved up and down for inserting the sample between the plates and making the metallic plate surfaces touch the dielectric sample surface.

(b) Probes Arrangement for Liquids

It consists of two polished brass cylinders fixed coaxially using threaded insulating gaskets at the two ends (Figure 2a). These gaskets have holes, in the lower one for allowing the experimental liquid to flow in between the cylinders, and in the upper one for communication with the atmosphere. Care has been taken to make the cylindrical electrodes of uniform cross section. This arrangement is mounted vertically and can be moved up and down with a rack-and-pinion set-up. It is put in a vessel containing the experimental liquid. The outer surface of the outer cylinder has a vertical scale to measure the height of the liquid with in the cylinders. Proper leads are provided for connection to the Capacitance Meter.

2. Samples

(a) Solids: Glass Plate, Bakelite Sheet, Teflon, PZT (Lead Zirconate Titanate)

(b) Carbon Tetrachloride (CCl_4) - supplied with the setup

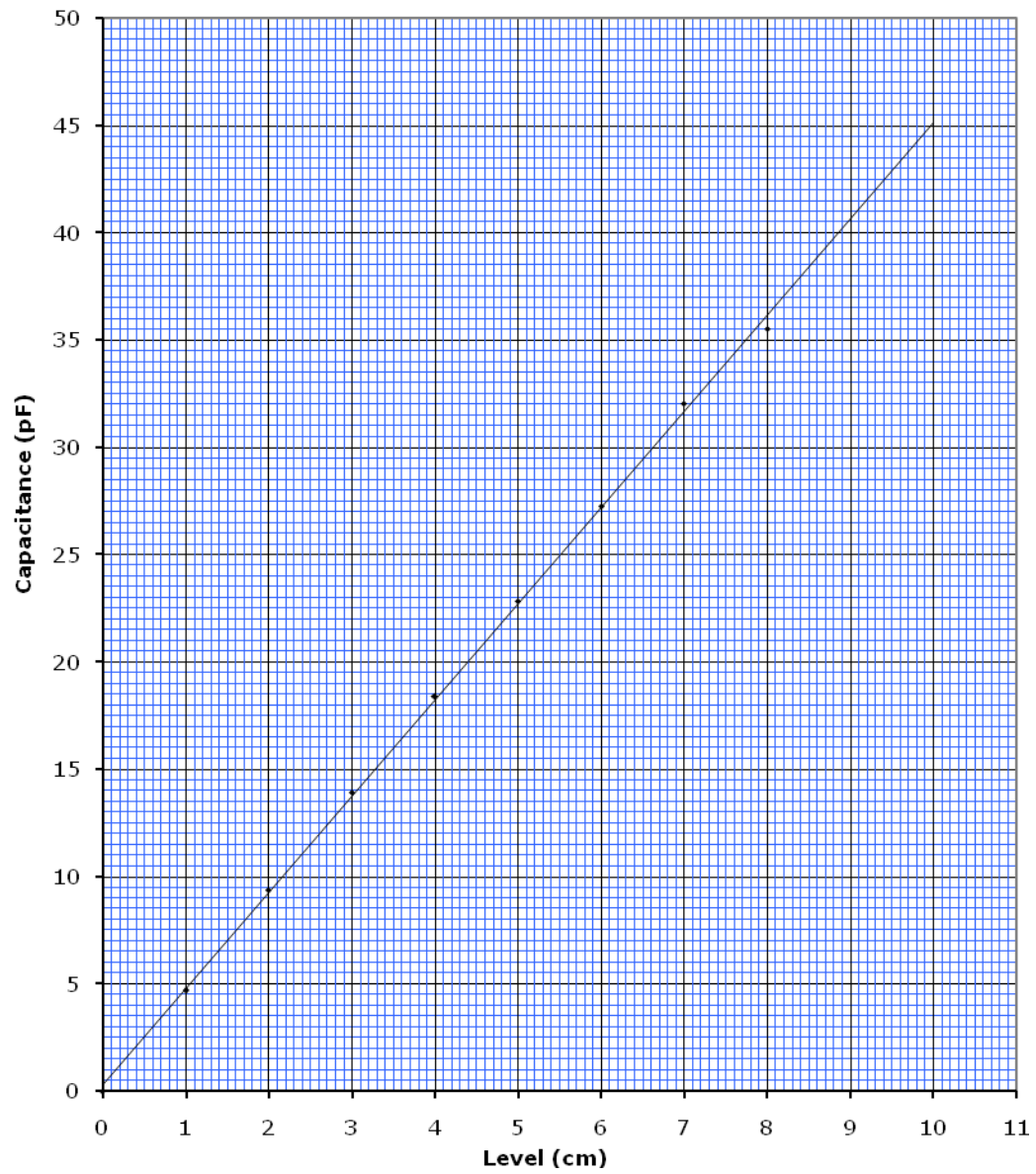
Other non conducting liquids such as acetone, ethyl alcohol, methanol, glycerol etc. can also be used for measurement.

3. Digital Capacitance Meter

This is compact direct reading microcontroller based high resolution instrument for the measurement of capacitance of the sample. It has facility of auto-zeroing.

Specifications	
Range	: 0 pf – 50 μf auto range
Resolution	: 0.01 pf
Display	: 16 x 2 LCD display with back light
Accuracy	: Better than 1%
Zero setting	: Push button zero setting

Dilelectric Constant of Solid & Liquids, Model DSL-01
Sample: Carbon Tetrachloride (CCl₄)



EXPERIMENTAL PROCEDURE

SETUP:

Choose the appropriate Sample Holder, as per sample size and connect the same to the Digital Capacitance Meter. Now add a little space among two plates of the sample holder and switch the capacitance meter 'ON' without keeping any sample. Press "ZERO ADJ" switch for 2-3 seconds, until the meter shows "Zero Cal". The capacitance meter is calibrated now.

SOLID SAMPLE:

The upper plate of the appropriate parallel plate capacitor of the arrangement for solids is raised by turning the top screw anticlockwise to keep/ insert the sample. The plate is then lowered by turning the top screw clockwise till the capacitor plates touch the sample surface. Switch the Meter "ON" and record reading.

LIQUID SAMPLE:

The glass container is put on the stand below the Coaxial Cylinder Set-up (CCS) as shown in Figure 2b. The container is filled with the experimental liquid up to a height of about 12 cm. The CCS is now lowered until the liquid touches the zero mark on the scale. The Capacity Meter is now adjusted to zero value by pushing zero set button for about a second (pressing it for longer duration will automatically set it to Auto Calibration mode, which is not desirable for this experiment). Remember, the result depends not on the actual value of capacitance, but on its variation with the height of the liquid. The CCS is then lowered in steps of 1 cm and the readings on the Capacity Meter are taken and recorded.

TYPICAL OBSERVATIONS, CALCULATIONS AND RESULTS

SOLID SAMPLE:

- (1) Room temperature: 26°C
- (2) Diameter of smaller capacitor plates: 10.0mm
- (3) Diameter of bigger capacitor plates: 50.0mm
- (4) Thickness of lead zirconate titanate Pellet: 2.55mm
- (5) Thickness of bakelite sheet sample = 1.5mm
- (6) Thickness of glass plate sample = 1.5mm
- (7) Thickness of teflon sample = 1.0mm

Capacity Measurement

Sr. No.	Sample	C (pF)
1	PZT (Lead Zirconate Titanate)	909.15
2	Glass	58.57
3	Bakelite	46.97
4	Teflon	32.28

$$\kappa_{\text{Bak}} = \frac{(C d)}{\epsilon_0 A} = \frac{46.97 \times 10^{-12} \times 1.5 \times 10^{-3}}{8.854 \times 10^{-12} \times 625 \times 3.141 \times 10^{-6}} = 4.053$$

$$\kappa_{\text{PZT}} = \frac{(C d)}{\epsilon_0 A} = \frac{909.15 \times 10^{-12} \times 2.50 \times 10^{-3}}{8.854 \times 10^{-12} \times 25 \times 3.141 \times 10^{-6}} = 3269.1$$

$$\kappa_g = \frac{(C d)}{\epsilon_0 A} = \frac{58.57 \times 10^{-12} \times 1.5 \times 10^{-3}}{8.854 \times 10^{-12} \times 625 \times 3.141 \times 10^{-6}} = 5.055$$

$$\kappa_t = \frac{(C d)}{\epsilon_0 A} = \frac{32.28 \times 10^{-12} \times 1.00 \times 10^{-3}}{8.854 \times 10^{-12} \times 25 \times 3.141 \times 10^{-6}} = 1.86$$

Liquid Sample:

Sample: Carbon Tetrachloride (CCl₄)

Dielectric Constant of free air (κ_0): 1.0059

External radius of the inner cylinder: 25.4 mm

Internal radius of the outer cylinder: 30.6 mm

Capacity Measurement:

S.No.	Liquid Level (cm)	Capacity (pf)
1	0.0	0.00
2	1.0	4.69
3	2.0	9.38
4	3.0	13.93
5	4.0	18.41
6	5.0	22.86
7	6.0	27.27
8	7.0	32.07
9	8.0	35.52

1. A plot of Capacitance v/s Liquid height is shown in Fig. 3.
2. The graph is linear. Its slope dC/dh is measured. For the present readings it comes out to be $3.67 \text{ pf/cm} = 3.67 \times 10^{-10} \text{ f/m}$.

Using Eq.(4),

$$\begin{aligned}\kappa - \kappa_0 &= \frac{(dC/dh)}{2\pi\epsilon_0} \ln(r_2/r_1) \\ &= \frac{4.444 \times 10^{-10} \times \ln\left(\frac{30.0}{25.4}\right)}{2 \times 3.141 \times 8.854 \times 10^{-12}} = \frac{4.444 \times 10^{-10}}{55.60 \times 10^{-12}} \times 0.166 = 1.327 \\ \kappa &= \kappa_0 + 1.327 = 2.327\end{aligned}$$

PRECAUTIONS AND SOURCES OF ERROR

1. The connecting leads internal capacitance through compensated by Zero Setting, but still could introduce some error.
2. Presence of vapour over the surface of the liquid.

REFERENCE

1. Principles of Electricity, L. Page and N.I. Adams, S. Chand & Co.(1955).
2. Classical Electricity and Magnetism, W.K.H. Panofsky and M. Phillips, Addison-Wesley (1962).
3. Fundamentals of Electromagnetics, M.A. Wazed Miah, Tata McGraw-Hill (1982).

TECHNICAL SUPPORT

Feed Back

If you have any comments or suggestions about this product or this manual please let us know. **SES Instruments Pvt. Ltd.** appreciates any customer feedback. Your input helps us evaluate and improve our product.

To reach SES Instruments Pvt. Ltd.

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Contacting for Technical Support

Before you call the SES Instruments Pvt. Ltd. Technical Support staff it would be helpful to prepare the following information:

- If you problem is with the SES Instruments Pvt. Ltd apparatus, note :
 - Model number and S. No (usually listed on the label at the backside of instrument).
 - Approximate age of the apparatus.
 - A detailed description of the problem/ sequences of events may please be sent by email or Fax.
- If your problem relates to the instruction manual, note;
Model number and Revision (listed by month and year on the front cover).
Have the manual at hand to discuss your questions.